HUM: Hear, Understand, Match for Unseen Audio-Text Alignment

Afeef Ali (Team Leader)
PID: aali11
Dept. of Computer Science
Virginia Tech
aali11@vt.edu

Kean Jaldin Guzman
PID: keanjg28
Dept. of Computer Science
Virginia Tech
keanjg28@vt.edu

Erdem Yilmaz
PID: erdemy
Dept. of Computer Science
Virginia Tech
erdemy@vt.edu

Anthony Merino
PID: merino
Dept. of Computer Science
Virginia Tech
merino@vt.edu

Project web page: https://aliai11.github.io/HUM/

***Abstract*—Automatic metrics for text-to-audio generation still agree poorly with human judgments of whether a clip actually matches its prompt. The ICASSP 2027 XACLE Grand Challenge asks for a model that predicts the mean human alignment score for an (audio, prompt) pair, and its 2027 edition tests on audio from generation systems never seen in training. We introduce HUM (Hear, Understand, Match), a trainable network that hears a clip through frozen audio and CLAP encoders, understands it with a QLoRA-tuned Qwen2.5-Omni audio-language model prompted as a yes/no verifier, and matches both signals through a gating network and an ordinal score head. We target the unseen-generator shift with a within-prompt pairwise ranking loss and evaluate with a leave-one-generator-out protocol. At this checkpoint we reproduce the official baseline exactly (validation/test SRCC 0.5844/0.5648) and match it with a head trained on cached M2D-CLAP embeddings (0.5881/0.5668). Under leave-one-generator-out evaluation, the within-prompt loss raises held-out-generator SRCC by 0.012 when added to an MSE loss but costs 0.008 in-domain test SRCC; added to the official loss at a smaller weight, it keeps in-domain SRCC and adds 0.006. Neither gain is yet statistically significant. A zero-shot comparison selects Qwen2.5-Omni-7B (SRCC 0.495) over AuroLA-rerank-7B (0.476) as the verifier to fine-tune.**

***Index Terms*—Text-to-audio evaluation, audio-language models, learning to rank, domain generalization, parameter-efficient fine-tuning.**

I. Introduction

A. Motivation

Text-to-audio (TTA) models such as AudioLDM and Tango can turn a sentence into sound, but checking whether the sound actually matches the sentence still requires human listening tests, which are slow, costly, and hard to reproduce. The standard automatic proxy, CLAPScore, correlates weakly with human relevance ratings: on the RELATE dataset, LAION-CLAP reaches a Spearman rank correlation (SRCC) of only 0.280 [1], [2], and contrastive audio-text models struggle with event order and attribute binding [3]. A learned evaluator that agrees with listeners would make TTA benchmarking cheaper and could serve as a reward signal for better generators.

B. Problem Statement

Given generated audio a and its prompt t, we predict a score $\hat{y}$ whose ranking over a test set matches the mean human alignment rating $y \in [0, 10]$. XACLE ranks systems by SRCC, with linear correlation (LCC), Kendall's tau (KTAU), and mean-squared error (MSE) as tie-breaks [4]. The 2027 edition adds an *unseen-generation-system* setting: test audio comes from generators that are absent from training [5]. A model can therefore look strong on validation by learning generator-specific artifacts and still fail on the test set, which makes the task a domain-generalization problem rather than only a regression problem.

C. Significance

Every 2026 system optimized correlation under seen generators. The 2026 baseline (BYOL-A + RoBERTa + LSTM) reached 0.3345 test SRCC [4], while the strongest systems reached 0.61–0.64 [6]–[8]. Since new TTA models appear every few months, a practical evaluator must generalize to generators it has never heard. At the same time, a zero-shot audio-language model judge already reaches 0.556 SRCC on RELATE [9], so strong in-domain correlation is an engineering target; generalization to unseen generators is the open research question.

D. Goals and Contributions

We separate engineering targets from the research bet:

- **G1.** Reproduce the official baseline (B1) and our CLAP-regression fallback (B4) by Checkpoint I (Oct. 20).
- **G2.** *Engineering:* reach in-domain validation SRCC at least equal to zero-shot AQAScore [9].
- **G3.** *Research bet:* reduce the leave-one-generator-out gap Δ (Eq. (7)) relative to B4, with a 95% bootstrap interval that excludes zero across at least three folds.
- **G4.** Submit a rule-compliant system to XACLE 2027 by Dec. 3.
- **G5.** Release a reproducible repository, a Gradio demo, and a demonstration video.

Our contributions are (i) a within-prompt ranking objective for generator-invariant alignment scoring, (ii) an unseen-generator benchmark protocol that reports Δ for re-implemented baselines, and (iii) a single-network judge that combines language-model verification with embedding similarity. Section V reports our progress at Checkpoint I.

II. Related Work

A. Embedding-Based and Learned Metrics

CLAPScore [10] measures how well audio and text match by comparing their CLAP embeddings. Human-CLAP improves CLAP by fine-tuning it on human ratings [1], while M2D-CLAP learns stronger general-purpose audio-language features [11]. RELATE released human relevance scores for TTA audio [2], and XACLE 2026 built its benchmark on this lineage [4].

The top XACLE 2026 systems extended these ideas. MoEScore gated several CLAP-based experts together with a BEATs [12]–RoBERTa cross-attention expert, reaching an SRCC of 0.6402 [6]. The third-place TMU system fine-tuned a compact audio-language model using CLAP pseudo-labels and a listwise ListNet loss [13], reaching 0.632 [7]. ListNet ranks clips across the whole batch regardless of prompt; HUM adds a ranking loss restricted to clips of the same prompt, so the model learns which clip best matches a prompt rather than which prompts are easier to score. SPO-CLAPScore standardized each listener's ratings, reaching 0.6142 [8].

HUM borrows gating, ranking, and listener standardization from these systems. Unlike them, HUM also targets generalization to audio-generation systems not seen during training, which is a problem none of these systems needed to solve, since XACLE 2026 tested only on seen generators.

B. Audio-Language Models as Judges

AQAScore asks Qwen2.5-Omni [14] whether a clip contains the events described in its prompt, using the probability of a "Yes" answer as the score; zero-shot, it reaches SRCC 0.556 on RELATE, above similarity-based metrics [9]. AuroLA repurposes Qwen2.5-Omni-7B for audio-text retrieval and adds a yes/no reranker [15]. HUM fine-tunes such a verifier with QLoRA [16], [17], rather than using it zero-shot, and fuses its output with an embedding-based signal instead of relying on the language model alone.

C. Generalization to Unseen Systems

Learned evaluators can overfit to generator-specific acoustic cues instead of semantic alignment [18]; in AudioMOS 2025, for example, systems degraded on unseen synthesis engines and vocoders [19]. Domain generalization methods counter this bias: domain-adversarial training with a Gradient Reversal Layer (GRL) [20] trains the model to score alignment without recognizing the source generator, and test-time adaptation methods like Tent [21] adjust predictions on unlabeled test batches. HUM's primary invariance signal, however, is the within-prompt ranking loss; the GRL branch and test-time calibration are add-ons that are kept only if they reduce Δ.

III. Proposed Approach

A. Overview

HUM is one network with three stages (Fig. 1). **Hear:** the clip passes through the frozen Qwen2.5-Omni audio encoder and frozen M2D-CLAP audio and text encoders, whose outputs are computed once and cached. **Understand:** the

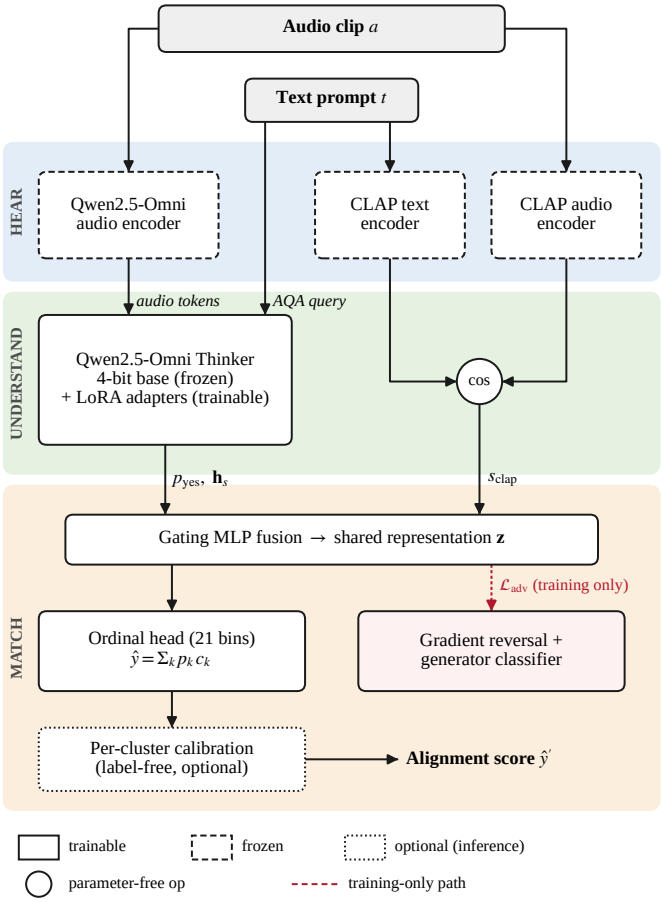


Fig. 1. HUM architecture. *Hear*: frozen audio and CLAP encoders. *Understand*: LoRA-tuned Qwen2.5-Omni verifier. *Match*: gated fusion, ordinal score head, and a training-only gradient-reversal generator classifier. Solid borders denote trainable modules; dashed borders denote frozen modules.

Qwen2.5-Omni Thinker, loaded with a 4-bit base and trainable LoRA adapters (the speech-output Talker is removed), receives the audio tokens and the query "Does this audio contain the sound events described by 't'?" It returns p_{yes}, the normalized probability of "Yes" versus "No," and $\mathbf{h}_s$, the hidden state at a dedicated score token. **Match:** a gating MLP fuses $\mathbf{h}_s$, p_{yes}, and the CLAP cosine similarity s_{clap} into a shared representation $\mathbf{z}$, and an ordinal head predicts probabilities p_k over $K = 21$ bins $c_k \in \{0, 0.5, \ldots, 10\}$. The score is the expected bin value:

$$\hat{y} = \sum_{k=1}^{K} p_k \, c_k. \tag{1}$$

The 2027 rules prohibit ensembling the results of multiple models but allow internal ensembling within a single model [5]; HUM is one network, so it is compliant.

B. Within-Prompt Ranking

The 2027 development set provides about three clips per prompt, each labeled with the system that generated it (Sec-

tion IV-B). For clips i and j of the same prompt with human scores $y_i > y_j$, we add a margin ranking loss:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \max\big(0,\ m - (\hat{y}_i - \hat{y}_j)\big), \tag{2}$$

where $\mathcal{P}$ is the set of within-prompt pairs whose score difference exceeds a noise threshold and m is a margin. Because both clips share the prompt, prompt difficulty cancels, and the model must rank clips by how well each one realizes the same text. We hypothesize that this reduces reliance on generator artifacts without requiring generator labels. Mini-batches are built so that all clips of a prompt appear together.

C. Generator Invariance

To filter generator-specific information out of $\mathbf{z}$, HUM adds an adversarial branch. During training, $\mathbf{z}$ passes through a GRL [20] into a two-layer MLP that predicts the source generator:

$$\mathcal{R}(\mathbf{z}) = \mathbf{z}, \qquad \frac{\partial \mathcal{R}(\mathbf{z})}{\partial \mathbf{z}} = -\mathbf{I}. \tag{3}$$

The 2027 release includes generator IDs, so the branch trains with a standard cross-entropy loss $\mathcal{L}_{\text{adv}}$ and no pseudo-domain clustering is needed. To keep adversarial gradients from destabilizing early training, its weight $\gamma(\tau)$ ramps from 0 to $\gamma_{\max}$ as in (5).

D. Training Objective

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \alpha\,\mathcal{L}_{\text{ord}} + \beta\,\mathcal{L}_{\text{mse}} + \delta\,\mathcal{L}_{\text{pair}} + \gamma(\tau)\,\mathcal{L}_{\text{adv}} \tag{4}$$

where $\mathcal{L}_{\text{rank}}$ is ListNet over the mini-batch [13], which directly rewards correct orderings and matches the rank-based SRCC; $\mathcal{L}_{\text{ord}}$ is the KL divergence from a Gaussian-softened target over the 21 bins ($\sigma = 0.5$), a soft-bin strategy used by top AudioMOS 2025 systems [19]; and $\mathcal{L}_{\text{mse}}$ on $\hat{y}$ stabilizes the score scale for the LCC and MSE tie-breaks. The weights α, β, δ, and m are tuned on validation data, and the adversarial weight follows

$$\gamma(\tau) = \gamma_{\max}\left(\frac{2}{1 + e^{-10\tau}} - 1\right), \tag{5}$$

where $\tau \in [0, 1]$ is training progress. The default system sets $\gamma_{\max} = 0$; with only eight generators the adversarial signal may be weak, so the GRL branch is kept only if it passes the acceptance rule in Section III-G.

E. Training Stages

S0 caches the frozen audio-encoder and CLAP outputs. **S2**, the default system, fine-tunes on XACLE with $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{ord}}$, $\mathcal{L}_{\text{mse}}$, and $\mathcal{L}_{\text{pair}}$. Three add-ons run only if they pass the acceptance rule: **S1**, a warm-up on declared external caption–audio pairs labeled by a CLAP teacher, which was the main performance driver for TMU [7]; **S3**, which enables the GRL branch; and per-listener score standardization [8], now possible because listener IDs are released.

F. Optional Test-Time Calibration

Audio from unseen generators can shift scores systematically. Without using test labels, we cluster the evaluation audio representations into C groups with K-means and adjust each score by its cluster's mean prediction μ_c relative to the overall mean μ:

$$\hat{y}' = \hat{y} - \lambda\,(\mu_c - \mu), \qquad \lambda \in [0, 1]. \tag{6}$$

A single monotonic map cannot change SRCC, so the correction must act per cluster. Because full equalization could erase genuine quality differences between generators, λ defaults to 0 and is tuned on validation LOGO folds. The step keeps model weights frozen and is applied only if the rules allow transductive use of the evaluation audio.

G. Baselines, Metrics, and Evaluation Protocol

Baselines: B1 is the official 2027 baseline, a frozen M2D-CLAP encoder with a trained MLP head [22]; B2 is zero-shot CLAPScore; B3 is zero-shot AQAScore [9]; and B4 is the B1 head trained on cached M2D-CLAP embeddings with our losses, which is our guaranteed submission. **Metrics:** we report SRCC (primary), LCC, KTAU, MSE, and per-generator SRCC, with 95% bootstrap intervals from 1,000 prompt-level resamples, because 2026 placements were separated by as little as 0.0001 SRCC [8]. **Protocol:** clips of one prompt never straddle training and evaluation. Leave-one-generator-out (LOGO) cross-validation trains on all but one generator and tests on the held-out one. The headline number is

$$\Delta = \text{SRCC}_{\text{seen}} - \text{SRCC}_{\text{unseen}}. \tag{7}$$

We report Δ for B2–B4 and HUM, which forms our benchmark contribution. **Acceptance rule:** a component is kept only if it lowers Δ without dropping in-domain SRCC below B4. **Ablations:** each loss term removed; with and without S1, GRL, calibration, and listener standardization; zero-shot versus fine-tuned verifier; and 3B versus 7B.

IV. System Design

A. Architecture and Workflow

The pipeline has seven steps: (1) ingest XACLE and declared external data, validate counts and score ranges, and remove RELATE clips that duplicate XACLE audio or prompts; (2) cache frozen features and build prompt-disjoint LOGO folds; (3) train B4, then HUM; (4) score every run with one shared metrics library; (5) select the final configuration with the acceptance rule; (6) run frozen inference on the evaluation release; and (7) export predictions in the official format. Every result comes from a notebook with fixed seeds.

B. Data Sets

Our primary data is the XACLE 2027 development set, described in Section V-C and Table II. RELATE [2] is our external dataset, and AudioCaps and Clotho provide caption–audio pairs for the optional S1 stage; all are on the organizers' pre-approved list.

C. Languages, Tools, and Compute

We build HUM using Python, PyTorch, Hugging Face Transformers and PEFT [16], bitsandbytes [17], torchaudio, scikit-learn, SciPy, Gradio, and GitHub. All compute runs on Google Colab Pro, with notebooks and results shared through one team Drive folder. Because the encoders are frozen and cached, all embedding experiments, including the 96 LOGO runs, run on CPU or an L4 GPU. A100 time is reserved for the 7B verifier, which we fine-tune in bf16 with gradient checkpointing, a batch size of 2–4 with gradient accumulation, and checkpoints saved every 200 steps so training can resume after session limits. We budget about 60 A100-hours. Qwen2.5-Omni (Apache-2.0), including the 3B fallback, is pre-approved.

D. Demo and Usability

A Gradio demo will take an audio file and a prompt and show $\hat{y}$, the 21 bin probabilities, the gating weights, and raw versus calibrated scores; we will report its GPU memory and latency with 4-bit quantization.

E. Risks and Fallbacks

If 7B training is too slow, we fall back to the 3B model or submit B4 alone. If the pairwise loss does not reduce Δ, we report a negative result with confidence intervals, which still supports the benchmark contribution. Section V shows that this risk is real.

V. Checkpoint I Progress Report

A. Status of Goals

G1 is complete: B1 is reproduced exactly, and B4 is trained and evaluated both in-domain and under LOGO. **G2** is met by B4 (SRCC 0.582 against 0.495 for our zero-shot AQAScore re-implementation on the same 498 validation clips) and remains to be shown for HUM. **G3 is not yet met:** the within-prompt loss reduces Δ in most LOGO folds, but most intervals include zero. **G4 and G5** are on schedule, although the first QLoRA run moved from Oct. 7–19 to Oct. 20 so that the zero-shot comparison could choose the backbone first.

B. Changes Since the Proposal

- The official baseline is now frozen M2D-CLAP with an MLP head [22] (formerly BYOL-A + RoBERTa + LSTM), essentially our planned B4, so B4 is now that head trained on cached embeddings.
- Generator and listener IDs are released, so LOGO folds use real labels and no pseudo-domain clustering is needed.
- Rules confirmed: training uses only the training split, ensembles of models are prohibited, up to four systems may be submitted, and all our models and data are pre-approved.
- We added AuroLA-rerank-7B [15] as a candidate verifier and moved all compute to Google Colab.

C. Data and Evaluation Setup

The development set has 7,500/3,000/3,000 training/validation/test pairs of 10-s, 16-kHz clips, rated by four listeners per pair (eight for test) from disjoint listener pools (2,323, 668, and 1,336 listeners). Clips come from real AudioCaps recordings and eight TTA generators (Table II). Our checks confirmed the row counts, that all ratings lie in $[0, 10]$, and that the embedding index matches the metadata. All nine sources appear in all three splits, so the official test split contains no unseen generator. Twelve training prompts also appear in validation (11) or test (1); we remove their 45 training clips in the LOGO runs.

The main results are scored by one shared metrics library that matches the official `evaluate.py` to four decimals and adds prompt-level bootstrap intervals and per-generator SRCC; paired bootstraps are computed in the experiment code. Each LOGO fold holds out one TTA generator from training *and* from best-epoch selection; AudioCaps is never held out because it is real audio. Each fold is run with three seeds for each of the four B4 losses, giving 96 runs.

D. Experiments

B1 and the embedding cache. The released B1 checkpoint, run with the official scripts, reproduces the published numbers to every reported digit. Since its encoders are frozen, we cached the normalized 768-dimensional M2D-CLAP audio and text embeddings for all 13,500 clips (83 MB instead of 4.2 GB of audio). The B1 head applied to the cache reproduces the released predictions within rounding (maximum difference 0.005).

B4 graders. All use B1's head (linear 1536→1024→256→1, ReLU, dropout 0.3) on the concatenated embeddings, with batches of 12 clips made of whole prompts, so they differ only in the loss: $\mathcal{L}_{\text{mse}}$ or the official loss $\mathcal{L}_{\text{off}}$ (a class-balanced clipped MSE plus a contrastive term [22]), each alone or with $\mathcal{L}_{\text{pair}}$ ($m = 0.5$ points, 0.5-point threshold). $\mathcal{L}_{\text{pair}}$ uses $\delta = 1$ with $\mathcal{L}_{\text{mse}}$; because $\mathcal{L}_{\text{off}}$ is about 12 times smaller than $\mathcal{L}_{\text{mse}}$, its δ is chosen on validation from $\{0.03, 0.1, 0.3, 1\}$. Training uses Adam (learning rate 2×10^{-5}) for 20 epochs, keeps the epoch with the best validation SRCC, and uses seeds 0–2. A code review fixed two bugs in the first version (cross-prompt pairs and uncopied best-epoch weights); all results use the corrected code.

B3 verifiers. Following AQAScore, the score is $10\, p_{\text{yes}}$ from the next-token logits for the query in Section III-A, run in bf16 on one A100 over 498 validation clips from 164 random prompts. AuroLA uses its own reranking prompt.

B2. The score is the cosine similarity s_{clap} of the cached, L_2-normalized M2D-CLAP audio and text embeddings, mapped to the 0–10 scale by a least-squares fit on the training split ($\hat{y} = 29.52\, s_{\text{clap}} - 2.76$); the map changes only MSE.

E. Results

In-domain (Table I): the $\mathcal{L}_{\text{mse}}$ grader slightly exceeds B1 on both splits with very small seed variance, confirming the

TABLE I
IN-DOMAIN RESULTS (ALL GENERATORS SEEN IN TRAINING). B4 ROWS: MEAN ± STD OVER 3 SEEDS

	Validation				Test			
Model	SRCC↑	LCC↑	KTAU↑	MSE↓	SRCC↑	LCC↑	KTAU↑	MSE↓
B2: CLAP similarity	0.4270	0.4923	0.2954	4.319	0.4195	0.5123	0.2890	3.811
B1: official baseline (released)	0.5844	0.5995	0.4186	3.637	0.5648	0.6142	0.3997	**3.185**
B4: grader, $\mathcal{L}_{\text{mse}}$ only	**0.5881**±.001	**0.6054**±.001	**0.4211**±.001	3.645±.027	**0.5668**±.001	**0.6161**±.002	**0.4010**±.000	3.216±.016
B4: grader, $\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{pair}}$	0.5872±.001	0.6035±.002	0.4198±.001	**3.604**±.013	0.5584±.004	0.6053±.004	0.3945±.003	3.235±.023

TABLE II
SOURCES IN THE DEVELOPMENT SET

ID	Source	Train†	Val / Test
0	AudioCaps (real recordings)	2,485	1,000 each
1–8	AudioGen, AudioLDM, AudioLDM 2, Make-An-Audio 2, Stable Audio, Tango, Tango 2, TangoFlux	496–747 each	250 each
	Total	7,455	3,000 each

†After removing the 45 prompt-overlap clips.

TABLE III
LOGO RESULTS, MEAN OF 3 SEEDS (BASE: LOSS WITHOUT THE PAIRWISE TERM)

	Unseen SRCC↑		Δ ↓	
Held-out generator (test)	Base	$+\mathcal{L}_{\text{pair}}$	Base	$+\mathcal{L}_{\text{pair}}$
Base $\mathcal{L}_{\text{mse}}$, $\delta = 1$				
AudioGen	0.499	0.498	0.061	0.058
AudioLDM	0.431	0.451	0.096	0.071
AudioLDM 2	0.678	0.671	−0.143	−0.141
Make-An-Audio 2	0.373	0.367	0.206	0.202
Stable Audio	0.514	0.535	0.042	0.015
Tango	0.563	0.582	−0.018	−0.044
Tango 2	0.477	0.496	0.093	0.064
TangoFlux	0.227	0.257	0.358	0.313
Mean, test	0.470	**0.482**	0.087	**0.067**
Mean, validation	0.437	**0.447**	0.143	**0.131**
Base $\mathcal{L}_{\text{off}}$, $\delta = 0.03$				
Mean, test	0.469	**0.475**	0.087	**0.081**
Mean, validation	0.433	**0.440**	0.145	**0.141**

cache and training pipeline, and $\mathcal{L}_{\text{off}}$ is nearly identical (validation/test SRCC 0.5867/0.5644; paired differences -0.004 to $+0.001$). Adding $\mathcal{L}_{\text{pair}}$ to $\mathcal{L}_{\text{mse}}$ ties it on validation (all paired intervals include zero) but lowers test SRCC by 0.008, significantly for two of three seeds. With $\mathcal{L}_{\text{off}}$, $\mathcal{L}_{\text{pair}}$ at $\delta = 1$ dominates the smaller loss (0.5428/0.5324), and $\delta = 0.3$ and 0.1 still hurt (0.5698/0.5450 and 0.5855/0.5585). The chosen $\delta = 0.03$ is neutral (0.5889/0.5631): one seed is significantly better on validation and one significantly worse on test.

Zero-shot verifiers: on the 498-clip subset, Qwen2.5-Omni-7B reaches SRCC 0.495, AuroLA-rerank-7B 0.476, and the trained B1 0.570. We choose Qwen2.5-Omni-7B because it scored higher, although the 0.019 difference is small for 164 prompts. The 0.08 gap to the trained model is what fine-tuning must close.

Unseen generators (Table III): generalization varies widely. TangoFlux and Make-An-Audio 2 are hardest when unseen (0.23 and 0.37), while AudioLDM 2 and Tango are easier than the pooled seen set, giving negative Δ. With $\mathcal{L}_{\text{mse}}$, $\mathcal{L}_{\text{pair}}$ raises unseen SRCC by 0.012 on test and 0.010 on validation and reduces Δ in 7 of 8 test folds and all 8 validation folds, but part of this comes from a lower seen SRCC (0.557 to 0.549 on test). With $\mathcal{L}_{\text{off}}$ at $\delta = 0.03$, the gain is half as large (0.006 on both splits; Δ falls in 6 of 8 folds on each), and seen SRCC is unchanged (0.556 on test). Few paired intervals exclude zero: 4 of 24 test fold-seed runs improve with $\mathcal{L}_{\text{mse}}$, and 2 improve and 1 worsens with $\mathcal{L}_{\text{off}}$, so G3 is not met. Measured against the mean within-generator SRCC instead, the reduction vanishes in both cases ($\leq$0.001).

Interpretation: on frozen embeddings, $\mathcal{L}_{\text{pair}}$ slightly improves the ordering within every generator, seen or unseen, rather than closing the gap. It fails the acceptance rule with $\mathcal{L}_{\text{mse}}$ at $\delta = 1$ but passes narrowly with $\mathcal{L}_{\text{off}}$ at $\delta = 0.03$, so B4 keeps that setting. A frozen-encoder head has little capacity to reorder clips of the same prompt, so we will re-tune δ and test the loss inside the fine-tuned verifier.

F. Progress by Team Member

Afeef Ali (leader). *Done:* coordination and team runbook; downloaded the data in Colab; reproduced B1; built and verified the embedding cache; ran and compared the zero-shot verifiers; reviewed and corrected the B4 notebooks and added the official-loss graders and δ sweep; drafted this report. *Next:* QLoRA fine-tuning of the Qwen2.5-Omni-7B Thinker with the score token, ordinal head, and Eq. (4).

Kean Jaldin Guzman. *Done:* data validation; the shared metrics library (official metrics, prompt-level bootstrap, per-generator SRCC), verified against `evaluate.py`; the automatic results table. *Next:* RELATE ingestion and deduplication, the ablation runner, and the submission formatter.

Erdem Yilmaz. *Done:* generator analysis; the LOGO protocol with prompt-disjoint training and hidden-generator-free model selection; the 96 LOGO runs and paired-bootstrap analysis; the B2 baseline. *Next:* the GRL branch (S3) and test-time calibration, evaluated with LOGO.

Anthony Merino. *Done:* the first B4 grader with the within-prompt ranking loss and prompt-grouped batch sampler. *Next:* caching Human-CLAP embeddings, the gating fusion MLP, and listener standardization.

G. Next Steps

The first QLoRA run starts Oct. 20; if no 7B epoch finishes by Oct. 26, we switch to the 3B model. Checkpoint II

(Nov. 10) will report the fused model under LOGO with the ablations above. The updated schedule is in Appendix B.

ACKNOWLEDGMENT

AI tools were used to help draft and edit this report and to assist with coding; all technical choices, results, claims, and citations were reviewed by the team.

REFERENCES

[1] T. Takano *et al.*, "Human-CLAP: Human-perception-based contrastive language-audio pretraining," in *Proc. APSIPA ASC*, 2025.
[2] Y. Kanamori, Y. Okamoto, T. Takano, S. Takamichi, Y. Saito, and H. Saruwatari, "RELATE: Subjective evaluation dataset for automatic evaluation of relevance between text and audio," in *Proc. Interspeech*, 2025, pp. 3155–3159.
[3] S. Ghosh *et al.*, "CompA: Addressing the gap in compositional reasoning in audio-language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
[4] Y. Okamoto, R. Takizawa, M. Kishi, Y. Kanamori, N. Tonami, R. Nagase, S. Takamichi, and K. Imoto, "XACLE Challenge 2026: The first x-to-audio alignment challenge," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2026, pp. 21877–21879.
[5] XACLE Organizers, "XACLE Challenge 2027," ICASSP 2027 Signal Processing Grand Challenge, 2026. [Online]. Available: https://xacle.org/2027/
[6] B. Sun, Y. Xiao, and H. Yin, "MoEScore: Mixture-of-experts-based text-audio relevance score prediction for text-to-audio system evaluation," in *Proc. IEEE ICASSP*, 2026, arXiv:2601.06829.
[7] A. Tsutsumi, K. Tanaka, and S. Shiota, "The TMU system for the XACLE challenge: Training large audio language models with CLAP pseudo-labels," in *Proc. IEEE ICASSP*, 2026, arXiv:2602.00604.
[8] T. Takano *et al.*, "SPO-CLAPScore: Enhancing CLAP-based alignment prediction system with standardize preference optimization, for the first XACLE challenge," arXiv:2601.02900, 2026.
[9] C.-Y. Kuan, K.-W. Chang, and H.-Y. Lee, "AQAScore: Evaluating semantic alignment in text-to-audio generation via audio question answering," arXiv:2601.14728, 2026.
[10] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE ICASSP*, 2023.
[11] D. Niizumi, D. Takeuchi, M. Yasuda, B. T. Nguyen, Y. Ohishi, and N. Harada, "M2D-CLAP: Exploring general-purpose audio-language representations beyond CLAP," *IEEE Access*, vol. 13, 2025.
[12] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 5178–5193.
[13] Z. Cao, T. Qin, T.-Y. Liu, M.-F. Tsai, and H. Li, "Learning to rank: From pairwise approach to listwise approach," in *Proc. ICML*, 2007, pp. 129–136.
[14] J. Xu *et al.*, "Qwen2.5-Omni technical report," arXiv:2503.20215, 2025.
[15] J. Xu, C. Thomé, D. Horak, W. Xie, and A. Zisserman, "Scaling audio-text retrieval with multimodal large language models," arXiv:2602.18010, 2026.
[16] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
[17] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
[18] K.-T. Huang, C.-C. Wang, C.-Y. Yang, H.-S. Lee, H.-M. Wang, and B. Chen, "Robust generative audio quality assessment: Disentangling quality from spurious correlations," arXiv:2603.16201, 2026.
[19] W.-C. Huang *et al.*, "The AudioMOS Challenge 2025," in *Proc. IEEE Autom. Speech Recognit. Understanding Workshop (ASRU)*, 2025, arXiv:2509.01336.
[20] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.
[21] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Proc. ICLR*, 2021.
[22] XACLE Organizers, "XACLE 2027 baseline model," GitHub, 2026. [Online]. Available: https://github.com/XACLE-Challenge/XACLE_2027_baselinemodel

APPENDIX A
TASK ASSIGNMENT FOR EACH MEMBER

TABLE IV
TASK ASSIGNMENT FOR EACH MEMBER (UPDATED AT CHECKPOINT I)

Member	**Primary Responsibilities**
[Leader] Afeef Ali	Leading overall coordination, data download, and the frozen-feature cache. Setting up the Qwen2.5-Omni Thinker with QLoRA: the AQA prompt and score token, the ordinal head, the Eq. (4) loss function, and Stage S2 training.
Anthony Merino	Building the CLAP-regression fallback (B4) and its loss-weight sweep. Also developing the gating fusion MLP, additional encoder caches (Human-CLAP), an optional S1 pseudo-label warm-up, and listener standardization.
Erdem Yilmaz	Generator analysis and the LOGO protocol, building the GRL head with its γ schedule for Stage S3, and handling per-cluster calibration and λ tuning.
Kean Jaldin Guzman	Managing data ingestion, validation, and RELATE de-duplication. Maintaining the metrics and bootstrap library, the results table, and the ablation runner, plus formatting the final submission.
All	Weekly syncs, code review of every notebook, and presentations.

APPENDIX B
SCHEDULE

TABLE V
PROJECT SCHEDULE (FALL 2026), WITH STATUS AT CHECKPOINT I

Week (2026)	**Tasks**	**Milestone / Deliverable**	**Status**
Sep 15–22	Form team, elect leader; survey XACLE 2026 papers; draft proposal; set up repo and web page	**Sep 22: Proposal due**	Done
Sep 23–29	Sept. 24 data + baseline release: download; verify rules, metric, generator/listener IDs; data validation	Rules/data checklist	Done
Sep 30–Oct 6	Run official baseline (B1), zero-shot CLAPScore (B2) and AQAScore (B3); cache CLAP features; build LOGO folds	Baseline table	Done by Oct 19
Oct 7–19	Train fallback B4; first LOGO Δ numbers; zero-shot verifier comparison; official-loss graders and $\mathcal{L}_{\text{pair}}$ weight sweep	**Oct 20: Checkpoint I** (G1)	Done; first QLoRA run moved to Oct 20
Oct 20–Nov 2	First QLoRA run of HUM (S2; 7B, or 3B if no 7B epoch by Oct 26); $\mathcal{L}_{\text{pair}}$ weight in the verifier; Human-CLAP cache and gating fusion; GRL head (S3); optional S1	Fused model on LOGO	Planned
Nov 3–9	The first five ablations; calibration prototype with λ tuning; demo prototype	**Nov 10: Checkpoint II + 15-min talk**	Planned
Nov 10–18	Remaining ablations; freeze method; dry-run submission on validation	Frozen configuration	Planned
Nov 19–Dec 2	Nov. 19 evaluation release: frozen inference, optional calibration, sanity checks; draft system description	Final predictions	Planned
Dec 1–8	Submit to XACLE (Dec. 3); final presentation (Dec. 1/3/8); record demo video	**Dec 3: XACLE submission**	Planned
Dec 9–11	Final report, repo tag, video upload, peer evaluations	**Dec 11 (8 PM): Final report**	Planned

APPENDIX C
ACKNOWLEDGMENT

This report was prepared with the assistance of AI tools, which were used to help generate, edit, and organize content, check references, and assist with code. All technical choices, results, claims, citations, and code are reviewed and verified by the team, and the team takes full responsibility for the work.